From the figure, it is clear that the two lines make the angles 60° and 120° with the x-axis

$$\begin{aligned} \text{Let } m_1 \text{ be } \tan 60^\circ &= \sqrt{3} \text{ and} \\ m_2 \text{ be } \tan 120^\circ &= \tan(180^\circ - 60^\circ) = -\tan 60^\circ = -\sqrt{3} \\ m_1 = \sqrt{3}, m_2 &= -\sqrt{3} \text{ and } b = 7 \\ \text{Equations of lines are } y &= m_1 x + b \text{ and } y = m_2 x + b \\ y &= \sqrt{3}x + 7 \text{ and } y = -\sqrt{3}x + 7 \end{aligned}$$

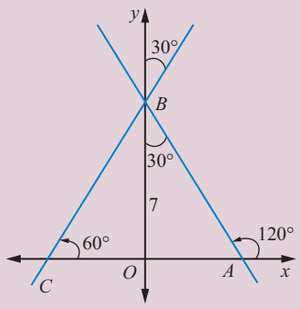


Figure 6.28

Whenever two points are given, one can apply two points form or point and slope form. Also when two intercepts are given, one can apply intercepts form or two points form The following example, is solved in chapter V, using the concepts of sequence and series. Let us solve this problem here, using the concepts of straight lines.

Example 6.12 The seventh term of an arithmetic progression is 30 and tenth term is 21.

(i) Find the first three terms of an A.P.
(ii) Which term of the A.P. is zero (if exists)
(iii) Find the relation ship between Slope of the straight line and common difference of A.P.

Solution:

Since there is a constant increase or decrease in arithmetic progression, it is a linear function.
Let the x-axis be the number of the term and the y-axis be the value of the term.
Let (x_1, y_1) and (x_2, y_2) be (7,30) and (10,21) respectively, using the equation

$$\begin{aligned} y - y_1 &= m(x - x_1) \text{ we get} \\ y - 30 &= \frac{21 - 30}{10 - 7}(x - 7) \\ y &= -3x + 51 \end{aligned} \tag{6.14}$$

(i) Substituting $x = 1, 2$ and 3 in the equation (**??**) we get the first three terms of AP as 48, 45, and 42.
(ii) Substituting $y = 0$ in equation (**??**) we get

$$0 = -3x + 51 \quad \Rightarrow x = 17.$$

That is seventeenth term of A.P. is zero.
(iii) clearly the slope of the line -3 is equal to the common difference A.P.

From this example, slope of the line is equal to common difference of A.P. (Try to prove it)